01 · LESSON

Environmental variation

9A Genetics and evolution · Lesson 1 of 7

Date _____ Class _____

Today I can

- ☐ I can describe abiotic and biotic environmental factors and give examples of environmental variation.
- ☐ I can tell continuous variation from discontinuous variation, and explain why environmental variation can make classification difficult.

What happens today

1. Do Now: the dragon of Krakow
2. Learn: Environmental factors · Environmental variation · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 5 to 7 · Workbook pages 3 to 4

What I learned today (one or two sentences)

02 · KEY VOCABULARY

Key Vocabulary

Term	Definition
species	a group of organisms that can breed together to produce offspring that can also breed
extinct	a species with no living members left
environment	the surroundings of an organism
abiotic factor	a physical (non-living) part of the environment, e.g. temperature
biotic factor	the activities of other organisms, e.g. predation, disease
characteristic	a feature of an organism
variation	the differences between the characteristics of organisms
environmental variation	variation caused by an organism's environment
continuous variation	variation that can have any value between two points
discontinuous variation	variation with only a limited set of possible values
classification	sorting organisms into groups

03 · QUICK REVIEW

Quick Review

the dragon of Krakow

A story says these bones are from a fire-breathing dragon killed by Krak, a shoemaker's apprentice.

1 What animals could the bones really be from?

2 What does 'extinct' mean?

3 Name one species that is extinct.

04 · CORE

Core 1 of 4

Everyone does these. Answer on the card or under it.

1 Match each key word to its meaning. Draw a line.

species	the surroundings of an organism
extinct	a non-living factor, e.g. temperature or light
environment	a species with no living members left
abiotic factor	a group of organisms that can breed together to produce offspring that can also breed
biotic factor	the activities of other organisms, e.g. predation or disease

Core 2 of 4

Everyone does these. Answer on the card or under it.

2 Tick one box in each row to show whether the factor is abiotic or biotic.

Environmental factor	Abiotic	Biotic
temperature		
predation (being hunted)		
amount of light		
competition for food		
rainfall		
infectious disease		

Core 3 of 4

Everyone does these. Answer on the card or under it.

3 Twenty potatoes of the same variety were weighed. Their masses in grams were: 142, 118, 165, 151, 137, 174, 129, 158, 146, 183, 121, 160, 149, 133, 171, 155, 140, 127, 163, 148.

a) Complete the tally chart.

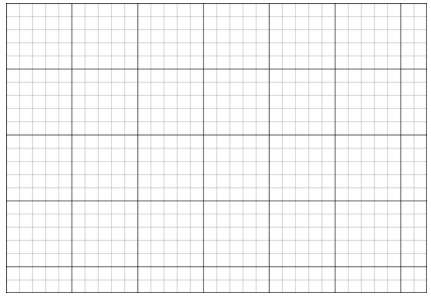
Mass range (g)	Tally	Number of potatoes
110 to 129		
130 to 149		
150 to 169		
170 to 189		

Core 4 of 4

Everyone does these. Answer on the card or under it.

3 *continued*

b) Plot a bar chart of the number of potatoes in each mass range on the grid below. Label both axes.



c) Is potato mass continuous or discontinuous variation? Give a reason.

Challenge 1 of 2

When Core is finished. Full sentences.

C1 A plant on a seashore has pink flowers and is 10 cm tall. On a nearby island, similar plants are over 50 cm tall with dark purple flowers. Scientists grew seeds from both places in the same laboratory. All the plants grew to look the same.

a) What does this tell you about the plants from the two areas?

b) Use this example to explain how environmental variation can make it difficult to identify plant species.

Challenge 2 of 2

When Core is finished. Full sentences.

C2 In 2003, bones of adult humans just over 1 m tall were found on Flores Island. Some scientists think they were our species but had too little iodine in their diet; others think they were a different species that is now extinct. Suggest what evidence would help scientists decide, and explain your reasoning.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 Give one abiotic factor and one biotic factor in a pond.

2 Is a scar an example of inherited or environmental variation?

3 Is foot length continuous or discontinuous variation? Give a reason.

Today I can: Goal 1 ☐ Goal 2 ☐